

Luther Bell

Software Engineer | Austin, TX

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Experience:

Software Engineer

General Motors — *October 2023 – Present*

Backend Development

- Modernized legacy Java and .NET backend systems by implementing security upgrades, enhancing data integrations, and improving service reliability.
- Upgraded a legacy backend service from 2-way TLS (mutual authentication) to OAuth 2.0, modernizing security and authentication flows.
- Built a backend feature to receive data from a new internal service, including developing SQL stored procedures to update and synchronize UI information.
- Diagnosed and resolved production-level backend issues, performing root cause analysis and implementing long-term fixes in Java/.NET services.
- Led migration of infrastructure from Windows Server 2012 to 2022 to ensure backend service continuity and improve system reliability.
- Built and maintained CI/CD pipelines using YAML, Azure DevOps, and GitHub Actions to automate and streamline deployments.
- Developed JMeter performance tests and containerized test environments with Docker to benchmark and optimize service scalability.
- Automated application installation on Windows servers using PowerShell scripts, eliminating manual setup and accelerating provisioning for new servers.
- Led multiple production deployments, coordinating across teams to ensure smooth releases with minimal downtime and quick issue resolution.
- Expanded cloud expertise by completing the AWS Fundamentals Specialization (Coursera), gaining hands-on experience with core AWS services including EC2, S3, IAM, RDS, and Lambda to strengthen backend engineering and cloud migration capabilities.

Database / SQL

- Created and updated SQL stored procedures to receive, process, and check incoming data for backend systems.
- Fixed production bugs by troubleshooting SQL queries and stored procedures, helping improve data accuracy and system stability.

Frontend Development

- Contributed to frontend enhancements using TypeScript to support new backend features and ensure seamless end-to-end user functionality.

Software Test Engineer (Backend and Frontend Automation)

General Motors — *March 2022 – October 2023*

- Developed Java-based automation frameworks using Selenium to validate backend functionality in service APIs.
- Automated over 300 regression test cases, significantly improving test coverage for backend microservices.
- Collaborated closely with developers to define acceptance criteria, reproduce complex bugs, and verify feature fixes pre-release.
- Integrated test automation with CI/CD pipelines to execute automation whenever needed and without disrupting daily tasks.
- Authored manual and automated test cases for backend APIs and frontend code in early development phases to reduce integration issues.

Education:

St. John's University - Queens, New York (May 2021)

Bachelor of Science - Computer Science

Certifications:

- AWS Fundamentals Specialization – Coursera (Completed Sept 2025)

Skills:

- **Languages:** Java, C#, Python, SQL, TypeScript, HTML/CSS
- **Frameworks/Libraries:** .NET, Selenium, Spring Boot, React
- **CI/CD & DevOps:** Azure DevOps, AWS (EC2, S3, IAM, RDS, Lambda), GitHub Actions, YAML, Docker, PowerShell
- **Testing:** JMeter, UI & API Automation, Manual & Automated QA
- **Databases:** Oracle, MS SQL Server, PostgreSQL
- **Tools:** Visual Studio, VS Code, Eclipse, IntelliJ, Git
- **Methodologies:** Agile, SAFe

Personal Projects:

NBA 2K Animation Database - In Progress

Tech Stack: React, TypeScript, Tailwind CSS, Spring Boot, Java, PostgreSQL, REST APIs, Docker (planned), Git

Built a full-stack app for NBA 2K players to find the best MyCareer animations based on build attributes like height, position, 3PT rating, and ball handle.

- **Backend:** Designed RESTful APIs using Spring Boot to serve filtered animation data and support future features like user voting and comments.
- **Frontend:** Built a responsive UI in React and Tailwind CSS with dynamic filtering and clean animation previews.
- **Infra/Planning:** Structured the codebase for future Docker containerization and scalable deployment.

NBA Player Stats Web Scraper

Tech Stack: Python, Flask, Next.js, TypeScript, Tailwind CSS, Pandas, Render, Vercel

Developed a full-stack app to scrape and display real-time NBA player stats.

- **Backend/Data:** Used Python with BeautifulSoup and Pandas to scrape and clean stats, exposed via Flask APIs with CORS support.
- **Frontend:** Built in Next.js and Tailwind CSS with a responsive layout, search filters, and loading states.
- **Deployment:** Deployed Flask API on Render and frontend on Vercel, with full CORS integration and mobile-friendly design.

AI Chatbot Web App

Tech Stack: Next.js, Flask, Hugging Face Transformers, Tailwind CSS

Built a chatbot interface with model-generated replies powered by open-source language models.

- **Backend:** Created a Flask API integrating Hugging Face transformers with prompt tuning and error handling.
- **Frontend:** Designed a chat UI in React/Next.js with loading states, conversation history, and clean styling using Tailwind CSS.

